UNIVERSITY OF WOLLONGONG AUSTRALIA

Tutorial 06

State models and cross-domain balances

Preparation: Lecture 06 modelling (slides 4–17) and state space (slides 19–26). The water-pressure question supplies its additional storage and flow laws.
Submission credit: 1% of the course grade for submitted work; correctness is not marked.
Two-session route: first practise modelling with Q1–Q4. In the second session, complete Q5 state matrices, eigenvalues and the transfer-function connection. The thermal and modal-response extensions are optional.

Question 1 Quick checks on physical states and output choice

Give a brief reason or calculation.
a) Energy stores. In an ideal series RLC circuit, which elements store energy and which dissipates it? Give a physical pair of independent state variables.

b) Matrix dimensions. A model has two states, one input and one output: $\dot{x}=Ax+Bu$, $y=Cx+Du$. Give the dimensions of A, B, C and D.

c) Instantaneous derivative. A normalized scalar state obeys $\dot{x}=-2x+3u$. At an instant when $x=1$ and $u=2$, find $\dot{x}$. Does this give the complete future response?

d) Output selection. Keep the series RLC state order $x=[v_C, i]^T$ and source-voltage input. If the output changes from capacitor voltage to series current, what changes in the state-space model?

Question 2 Complex impedance and a mechanical analogue

Use the series RLC circuit in **Figure 1** (Lecture 6, slides 8–10), with R = 4 Ω, L = 1 H and C = 0.25 F. Input u(t) is source voltage; output $v_C(t)$ is capacitor voltage. All initial states are zero.

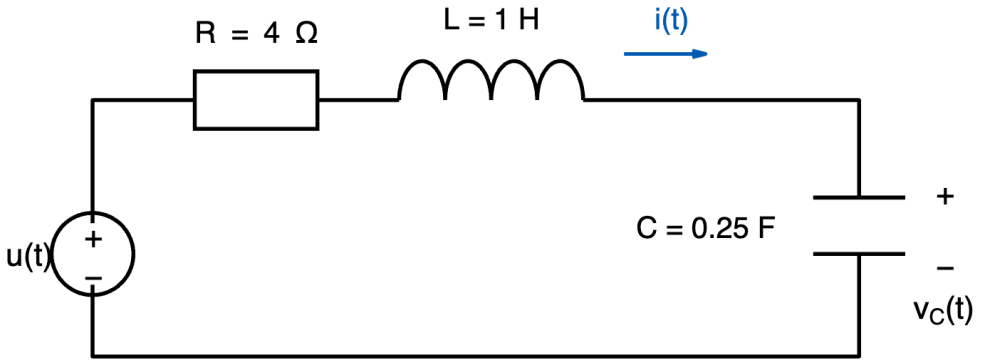


Figure 1. Series RLC circuit; source voltage u(t) and capacitor-voltage output vC(t).

Use the element impedances and voltage-divider relation below.

$$Z_R = R, Z_L = sL, Z_C = \frac{1}{sC}$$

$$G(s) = \frac{Z_C}{Z_R + Z_L + Z_C}$$

a) Substitute the impedances and parameters to obtain the capacitor-voltage transfer function G(s). Evaluate G(j2), then find its magnitude and phase.

b) Let q be capacitor charge and i its time derivative. Use the electrical and mechanical balances below. Match the corresponding coefficients and variables in the force–voltage analogy. What mechanical coefficients correspond numerically to L = 1, R = 4 and C = 0.25?

$$Lq'' + Rq' + \frac{q}{C} = u$$

$$mx'' + bx' + kx = F$$

Question 3 A single water tank

A tank has constant cross-sectional area A = 0.50 m² and linear outlet resistance R = 40 s/m². Work with small changes about a steady operating condition: h is the level change in metres, and q_i and q_o are inflow and outflow changes in m³/s. Assume the supplied linear laws remain valid over the stated changes. Initially h = 0.
Use the volume balance and linear outlet law below.

$$A\ dh/dt = q_i - q_o, \qquad q_o = h/R$$

For a first-order step response from zero initial deviation, use the formula below. At one time constant, the change is approximately 63.2% of its final value.

$$h(t) = h\infty[1 - \exp(-t/\tau)]$$

a) Substitute the outlet law into the volume balance. Insert the parameters to obtain the first-order ODE, then write the zero-state level-to-inflow transfer function $H(s)/Q_i(s)$.

b) Read the time constant and DC gain. Give the units of each. How many independent storage states does this model have?

c) The inflow increases by 0.010 m³/s and remains there. Find the final level change and its value at one time constant. State whether the level rises monotonically or oscillates.
MATLAB verification. Use ode45 for the 0.010 m³/s inflow step from zero level deviation. Overlay numerical and analytical responses in one labelled figure. Mark and report the level at one time constant; check the final level and monotonic rise. Base MATLAB is sufficient.

Question 4 A first-order water-pressure chamber

A water supply feeds a compliant chamber through a restriction. A flexible diaphragm or accumulator provides the effective storage; do not treat a rigid vessel of incompressible water as this model. The chamber has no outlet. Use linear resistance R = 20 000 kPa·s/m³ and effective volume compliance C = 0.00020 m³/kPa. All pressures are small changes from the initial operating pressure. Let p_i be the supply-pressure change, p the chamber-pressure change and q the inflow. Initially p = 0.
Use the flow and storage laws below.

$$q = (p_i - p)/R, \qquad C\ dp/dt = q$$

For a first-order step from zero initial pressure change, use:

$$p(t) = p\infty[1 - \exp(-t/\tau)]$$

a) Combine the two laws and insert the parameters. Write the first-order ODE and zero-state pressure transfer function $P(s)/P_i(s)$. Read its time constant and DC gain.

b) The supply-pressure change steps to 20 kPa and remains there. Find the final chamber-pressure change and its value at one time constant. What is the final inflow?

c) If the restriction resistance doubles while compliance stays unchanged, what happens to the time constant and final pressure? Give a brief physical explanation.

Question 5 State matrices eigenvalues and transfer function

Core state-space practice for the second Lecture 6 session.
A normalized second-order model is supplied below. Time is in seconds; z is normalized displacement, its rate has units s^{-1}, and u is normalized acceleration with units s^{-2}. The coefficients 5 and 6 have units s^{-1} and s^{-2} respectively.
$\ddot{z} + 5\dot{z} + 6z = u, x_1 = z, x_2 = \dot{z}, y = x_1$

Use the stated state order. Eigenvectors, modal trajectories and matrix exponentials belong to optional Extension B.
a) Write the two scalar state equations, then identify A, B, C and D in the state and output equations.
$\dot{x} = Ax + Bu, y = Cx + Du$

b) Find the eigenvalues from $\det(\lambda I - A)=0$. State whether free motion decays and which eigenvalue gives the slower decay.
c) With zero initial states, eliminate the states in the Laplace domain to obtain Y(s)/U(s). Compare its poles with the eigenvalues.

Extension A thermal target that may be unattainable

After class / optional extension

A well-mixed chamber has heat capacity 900 J/°C, thermal resistance 0.2°C/W to ambient at 20°C, and initial temperature 30°C. Heater power is held at 80 W. Its maximum available power is 100 W.

a) Write the heat balance. Find the time constant and equilibrium temperature for an 80 W heater. Use the first-order model to state whether temperature rises or falls from its initial value, and identify its limiting value.

b) Can any controller hold a target of 50°C using this heater? Calculate the required steady power and maximum achievable equilibrium.

Extension B Modal coefficients and state trajectories

Optional strengthening practice; the core Q5 does not require this calculation.
Use the Q5 model. The input steps to 1 s^{-2} from zero, and both initial states are zero. The eigenvectors and modal form are supplied below.

$$v_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$x(t) = x_{ss} + c_1 v_1 e^{\lambda_1 t} + c_2 v_2 e^{\lambda_2 t}$$

a) Find the equilibrium, determine the two modal coefficients and write both state trajectories. Check initial and final values.
b) Separately set u=0 with initial state [1, −2 s^{-1}]^T. Use the supplied slow eigenvector to find the free output. Explain why G(s)U(s) alone misses it.

Method reference

Quick-check sources: Reference/tutorials/Tute 4 Tutorial 4 Questions, Q2–Q3 (RLC laws and impedance); Ogata, Modern Control Engineering, 5th ed., Sec. 2–4, pp. 29–33, and Sec. 3–3, pp. 72–73. The matrix and output-selection checks are new short adaptations, not copied textbook exercises.
Method sources: Reference/tutorials/Tute 5/Tutorial 5 Solutions, Q1 (single-tank balance); Ogata, Modern Control Engineering, 5th ed., Section 4-2, pp. 103–104 (liquid level), Section 4-3, p. 109 (first-order pressure model), and Section 4-5 (thermal models). Q4 independently adapts the pressure resistance–storage method to a supplied effective volume compliance for a water accumulator; the textbook example is pneumatic. System Dynamics, 4th ed., Section 6-4 supports the mechanical–electrical analogy. Parameters are independently adapted. Page numbers refer to printed pages.
State-space practice source: Lecture 6, slides 19–24. Q5 covers matrices, eigenvalues and the transfer-function connection. Extension B supplies eigenvectors and retains the optional modal-response exercise. No controllability or observability test is required.